

AR 520: Design of Mechatronics Systems

Course Code: AR 520

Course Name: Design of Mechatronics Systems

L-T-P-C: 2-0-2-3

Prerequisites: IC 114: Linear Algebra

Intended for: UG, PG and Ph.D.

Mutual Exclusion: None

Approval: 48th Senate (in the B.Tech General Engineering curriculum); syllabus approved in 64th BoA (recoded from AR 503: Mechatronics)

Preamble

Students will be introduced to the main topics in the Mechatronics which blends the disciplines of mechanical engineering, computing, and electronics to create a huge range of electromechanical systems. It includes Introduction to the design of electromechanical systems. Interfacing sensors and actuators to a personal computer and a single-board computer. Electrical and mechanical design, prototyping, and construction. Dissection of a commercial mechatronic product. Students work in teams to produce final computer-controlled electromechanical projects of their own design.

Course Outline

Introduction to mechatronics, Example systems and architecture, Basic circuits review, Nonlinear circuit elements, Motor control. Various types of controllers. Data conversion devices. Review of programming. Number systems. Single-board computer. Digital design and integrated circuits. Sensors. Actuators, Power transmission, torque, speed, power equations, efficiency, inertia. Hydraulics, Pneumatics. Design and fabrication of Mechatronics systems. Final project.

Course Modules

- **Introduction to Mechatronics:** Introduction, Examples of Mechatronic systems, Measurement systems, Basic Electrical Elements (Resistor, Capacitor and Inductor), Kirchhoff's Laws, Voltage and current sources, meters, Equivalent circuits, Alternating Current Circuit Analysis and Impedance. (2 L)
- **Semiconductor Electronics:** Semiconductor physics for electronic devices, Diodes: Junction diode, Zener diode, Voltage regulators, Bipolar Junction Transistor-Switch and packages, Darlington transistor and Phototransistor & Optoisolator, Field-Effect Transistors. (4 L)
- **System response:** Amplitude linearity, Fourier Series Representation of Signals, Bandwidth and Frequency Response, Phase Linearity, Distortion of Signals, Dynamic Characteristics of Systems: Zero-Order System, First-Order System and Second-Order System, Frequency Response of a System and System Modeling and Analogies. (4 L)
- **Analog Signal Processing Using Operational Amplifiers:** Amplifiers, Operational Amplifiers: Ideal Model, Inverting, Noninverting Op Amps, Summer, Difference, Integrator, Differentiator, Comparator, The real operational Amplifier. (4 L)

- **Digital Circuits:** Digital Representations, Combinational Logic and Logic Classes, Timing Diagrams, Boolean Algebra, Design of Logic Networks, finding a Boolean Expression Given a Truth Table, Sequential Logic: Flip-Flops, Applications of Flip-Flops, TTL and CMOS Integrated Circuits, Using data sheets and standard symbols. (4 L)
- **Microcontroller Programming Interfacing and data acquisition:** Microprocessors and Microcomputers, Microcontrollers, The PIC Microcontroller, The Arduino Prototyping Platform, Interfacing, Serial Communication, Method to Design a Microcontroller-Based System, Prototyping and Design, Quantizing theory, Analog-to-Digital Conversion and Digital-to-Analog Conversion. (4 L)
- **Sensors:** Position and Speed Measurement, Stress and Strain Measurement, Temperature Measurement, Vibration and Acceleration Measurement, Pressure and Flow Measurement, Semiconductor Sensors and Microelectromechanical Devices. (3 L)
- **Actuators:** Electromagnetic Principles, Solenoids and Relays, Electric Motors, Stepper Motors, RC Servomotors, Selecting a Motor, Hydraulics and Pneumatics. (3 L)

Laboratory/Practical/Tutorial Modules

- i. Introduction to simulators for simple circuits and Arduino programming.
- ii. Introduction to practical considerations: Breadboard, Multimeter, Resistor circuits.
- iii. Function generators, Oscilloscope, Capacitors, RC Filters. Simulation in TinkerCAD and implementing hardware.
- iv. LEDs, Diodes, Zener Diodes, Rectification: Half wave rectification and Full wave rectification.
- v. Bipolar Junction Transistor as a switch and an amplifier. Photoelectric Circuit.
- vi. Operational Amplifiers hardware and Timer, 555 Timer as an oscillator.
- vii. Basics of soldering: Soldering wires and Soldering components to a perf board.
- viii. 3D Design and 3D Printing.
- ix. Sensors and Actuators I: Orientation sensor, Photocell and Distance sensor.
- x. Sensors and Actuators II: DC motor, Servo motor and Stepper motor.

Text Books

1. Alciatore, David G, *Introduction to Mechatronics and Measurement Systems*, McGraw Hill, 5th edition, 2019.
2. Bolton, William. *Mechatronics: electronic control systems in mechanical and electrical engineering*. Pearson Education, 7th edition, 2023.

References

1. Mahalik, Nitaigour Premchand. *Mechatronics*. Tata McGraw-Hill.
2. Bishop, Robert H. *Mechatronics: an introduction*. CRC Press.

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3. *Mechatronics system design* by Devdas Shetty and Richard A. Kolk, Cengage Learning.
 4. *Mechatronics* by G. Hegde, Jones and Bartlett.